1. What is a Database?

- A database is a giant, organized library
- Table = a shelf
- Row = one book
- Column = a label (title, author…)
- Netflix, shops and schools all use databases

```
SELECT * FROM books
LIMIT 3;
```

id	title	author	year	genre
1	Harry Potter	J.K. Rowling	1997	Fantasy
2	Matilda	Roald Dahl	1988	Comedy
3	Wonder	R.J. Palacio	2012	Drama

Anyone can learn this — easy!

2. What is SQL?

- SQL = Structured Query Language
- It is how we ask the librarian
- SELECT = "show me"
- * = every column
- Or list just the columns you need

```
SELECT title, author
FROM books LIMIT 4;
```

title	author
Harry Potter	J.K. Rowling
Matilda	Roald Dahl
Wonder	R.J. Palacio
The Hobbit	J.R.R. Tolkien

Tip: end each query with a semicolon ;

3. WHERE = a filter

- = equals (text goes in quotes)
- > greater than, < less than
- AND: both must be true
- OR: either one is enough
- LIKE 'H%': starts with H

```
SELECT title, year FROM books
WHERE genre = 'Fantasy'
AND year > 1990;
```

title	year
Harry Potter	1997
Coraline	2002

Only the books that pass the test come back

4. ORDER BY + LIMIT

- ORDER BY sorts the results
- ASC = small to big (default)
- DESC = big to small
- LIMIT 3 = only the top 3
- Order: SELECT, FROM, WHERE, ORDER BY, LIMIT

```
SELECT title, year FROM books
WHERE year < 2000
ORDER BY year ASC
LIMIT 2;
```

title	year
The Hobbit	1937
Charlotte's Web	1952

You just wrote a real SQL query!

5. Practice Free Online

- sqliteonline.com (choose SQLite)
- db-fiddle.com (choose SQLite)
- sqlime.org
- sqlbolt.com (free lessons)
- Paste the setup script from the guide, click Run

Coming up in Part 2

- COUNT, SUM, AVG: do math on data
- GROUP BY: sort into baskets
- HAVING: filter the baskets

Part 2 link: [PART 2 LINK]

Anyone can learn — easy! Subscribe for Part 2.